FSG-Net: Frequency-Spatial Synergistic Gated Network for High-Resolution Remote Sensing Change Detection

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***Abstract*—Change detection (CD) from high-resolution remote sensing (RS) images lies as a cornerstone of Earth observation applications, yet its efficacy is often compromised by two critical challenges. First, false alarms are prevalent as models misinterpret radiometric variations from temporal shifts (e.g., illumination, season) as genuine changes. Second, a nonnegligible semantic gap between deep abstract features and shallow detail-rich features tends to obstruct their effective fusion, culminating in poorly delineated boundaries. To step further in addressing these issues, we propose the frequency-spatial synergistic gated network (FSG-Net), a novel paradigm that aims to systematically disentangle semantic changes from nuisance variations. Specifically, FSG-Net first operates in the frequency domain, where a discrepancy-aware wavelet interaction module (DAWIM) adaptively mitigates pseudochanges by discerningly processing different frequency components. Subsequently, the refined features are enhanced in the spatial domain by a synergistic temporal–spatial attention module (STSAM), which amplifies the saliency of genuine change regions. To finally bridge the semantic gap, a lightweight gated fusion unit (LGFU) leverages high-level semantics to selectively gate and integrate crucial details from shallow layers. Comprehensive experiments on the CDD, GZ-CD, and LEVIR-CD benchmarks validate the superiority of FSG-Net, establishing a new state-of-the-art with F1-scores of 94.16%, 89.51%, and 91.27%, respectively. The code is available at https://github.com/zxXie-Air/FSG-Net.**

***Index Terms*—Change detection (CD), feature interaction, frequency-spatial analysis, gated fusion, remote sensing (RS).**

I. Introduction

CHANGE detection (CD) in remote sensing (RS) images is a vital technology for tracking geospatial changes [1], [2], with broad applications in urban planning [3], land monitoring [4], and disaster evaluation [5]. Still, the advent of vast submeter resolution datasets from sensors like WorldView, QuickBird, and Gaofen-2 has posed significant challenges to conventional methods like CVA [6] and Random Forest [7], revealing their diminished efficacy [8]. This has paved the way for deep learning-based CD to become the dominant paradigm, owing to the inherent capability of architectures like convolutional neural networks (CNNs) to automatically extract hierarchical features from large-scale, complex data [9]. Unfortunately, even these advanced deep learning models are not without their own set of critical obstacles.

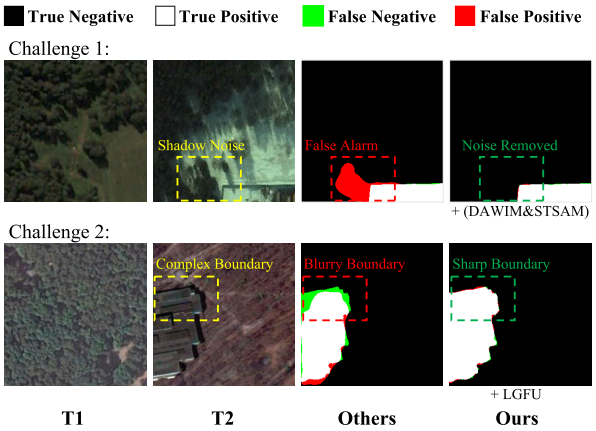


Fig. 1. Illustration of the two critical challenges in CD. Challenge 1: Pseudochanges caused by discrepancies in imaging conditions. Challenge 2: Imprecise boundary delineation resulting from the semantic gap between multilevel features.

Technically, the objective of CD tasks is to identify semantic changes of interest at a pixel-wise level from two co-registered RS images acquired at different times [10]. However, this process is hindered by two primary obstacles arising from the complex imaging environment. First, discrepancies in acquisition conditions—such as varying illumination angles and seasonal phenology—introduce significant pseudochanges. For instance, Challenge 1 in Fig. 1 illustrates how radiometric variations like strong shadows are visually prominent and are often misinterpreted by existing methods as genuine changes (false alarms). Second, a nonnegligible semantic gap exists between deep, abstract features and shallow, detail-rich features. This gap obstructs effective feature fusion, often leading to blurry

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boundaries [11]. In Challenge 2, for scale-sensitive objects like buildings, the lack of fine-grained details results in imprecise edge delineation. Therefore, to accurately identify changes in such complex scenarios, a trustworthy CD model is expected to embody the following two capabilities.

1) *Pseudochange Suppression:* Distinguishing genuine changes of interest from a multitude of task-irrelevant interference.
2) *Semantics-Aware Alignment and Fusion:* Ensuring a coherent fusion of multilevel features, aligning abstract semantics with fine-grained spatial details to produce well-delineated boundaries.

To mitigate background interference, techniques such as large-kernel convolutions and attention mechanisms have been widely adopted to incrementally expand receptive fields and amplify salient information [12], [13], [14], [15], [16], [17]. Concurrently, facilitating feature interaction between bi-temporal representations is considered paramount for robust change discrimination, where strategies like correlation computation explicitly model cross-temporal relationships [10], [18], [19]. However, purely spatial interactions inevitably operate on entangled pixel values where semantic changes and radiometric variations (e.g., illumination shifts) are inherently intertwined, obstructing the fundamental disentanglement of authentic changes from background noise [20]. In contrast, analyzing changes in the frequency domain offers a physics-aware perspective to decouple these variations. Specifically, low-frequency components capture the smooth, global characteristics of the image, encapsulating variations in illumination and background patterns [21]. Conversely, high-frequency components focus on regions of rapid intensity variation, such as edges and textures, which are critical for discerning fine-grained structural details and boundaries [22]. Driven by these insights, various techniques, including the discrete cosine transform [23] and fast Fourier transform [24], have been utilized to decouple object identity from stylistic variations [25], [26], [27]. Nevertheless, most existing frequency-based approaches often apply a rigid strategy across all frequency subbands, ignoring that distinct frequency components require specialized processing mechanisms [20]. Furthermore, they often fail to establish a synergistic framework to collaboratively leverage the complementary strengths of both frequency and spatial domains.

In deep neural networks, low-level features capture fine-grained spatial details, while high-level features encode abstract semantic context [10], [18]. A critical challenge, therefore, lies in the coherent integration of these disparate feature levels for CD tasks. Baseline approaches using simple concatenation or addition often fail to reconcile the semantic gap between deep, abstract concepts and shallow, fine-grained details, leading to blurred boundaries [19], [28]. To address this, recent studies have developed sophisticated bridge mechanisms to harmonize hierarchical features. For instance, hierarchical interaction strategies progressively fuse multiscale features in a coarse-to-fine manner [24], [29]. Similarly, alignment-based techniques, such as flow fields or deformable convolutions, have been employed to spatially calibrate features prior to fusion [30], [31]. While effective to some extent, these intricate interaction mechanisms often incur high computational complexity. Conversely, naive fusion without selective filtering risks propagating shallow background noise into the final prediction, thereby degrading detection precision.

Answering this call, we introduce FSG-Net, a frequency-spatial synergistic framework that separates the change signal from the nuisance and reconciles feature asymmetry in a principled manner. The core idea behind FSG-Net lies in: 1) adopting a synergistic frequency-spatial separation strategy to discern targets of interest and 2) leveraging deep semantics to refine multilevel feature details.

Concretely, we first introduce a discrepancy-aware wavelet interaction module (DAWIM) that operates on multifrequency wavelet subbands, where cross-temporal discrepancy cues drive adaptive subband interaction and reweighting. This yields a learnable frequency mask that attenuates low-frequency radiometric shifts while preserving high-frequency structural evidence, and the modulated representation is reprojected to the spatial stream through a residual path. Building on these refined features, a synergistic temporal–spatial attention module (STSAM) is proposed to couple the temporally enhanced cross-attention (CrossAtt) (to exchange long-range contextual cues between the two times) with coordinate attention (to encode positional priors), thereby amplifying saliency in truly changed regions and stabilizing local structures. Finally, to bridge the semantic gap between deep and shallow features—a common cause of blurred change boundaries—we propose a lightweight gated fusion unit (LGFU). This module generates semantics-driven gates from deep features to selectively propagate fine-grained details from earlier layers, yielding crisp boundary delineation with minimal computational overhead. To summarize, the main contributions of this work are as follows.

1) We propose the frequency-spatial synergistic gated network (FSG-Net), a novel architecture that pioneers a frequency-spatial synergistic pipeline to systematically disentangle semantic changes from nuisance variations in RS images.
2) To suppress background interference arising from discrepancies in acquisition conditions, we introduce a wavelet interaction module, termed DAWIM, that applies tailored strategies to different subbands in the frequency domain, effectively mitigating false alarms from radiometric shifts.
3) A novel spatial–temporal attention module STSAM is designed to synergistically amplify the saliency of genuine changes by simultaneously modeling global context and preserving fine-grained local details through coupling cross- and coordinate-attention (CoordAtt).
4) We devise a lightweight unit LGFU that bridges the deep-shallow semantic gap to sharp change boundaries. It features semantics-driven gates to selectively fuse features, yielding crisp boundary delineation with high computational efficiency.

The remainder of this article is organized as follows. Section II reviews related work in deep learning-based CD.

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Section III elaborates on the proposed FSG-Net, detailing the overall architecture and its core modules. Section IV describes the experimental setup, presents a comprehensive comparison and analysis against state-of-the-art methods, and includes ablation studies to validate our design choices. Finally, Section V concludes this article and discusses potential future work.

II. Related Work

A. Deep Learning-Based CD

Driven by the powerful feature representation of deep learning, CD methods have burgeoned in recent years, evolving from seminal architectures like FCN and ResNet [32], [33] to advanced models incorporating multiscale strategies and Transformers [34], [35], [36]. Based on their core processing paradigms, we categorize existing methods into three classes: CNN-based, Transformer-based, and the currently prevalent hybrid architectures.

Daudt et al. [37] first introduced a weight-sharing Siamese FCN framework to the CD task, presenting three variants that explored both early fusion (FC-EF) and feature-level fusion through concatenation (FC-Siam-Conc) and differencing (FC-Siam-Diff). Techniques such as large-kernel convolutions [12] and dilated convolutions [13] are frequently employed to expand the effective receptive field, thereby enabling CNN models to capture richer contextual information. Simultaneously, various attention mechanisms, including spatial [38] and channel attention [39] have been widely adopted to enhance the saliency of genuine change regions. Despite these enhancements, CNNs remain fundamentally constrained by their restricted local receptive fields, which prevent effective modeling of long-range semantic dependencies crucial for distinguishing genuine changes from environmental variations [40].

The Transformer architecture has achieved remarkable success in modeling sequential dependencies through its self-attention (SelfAtt) mechanism [41]. Breaking the constraints of CNNs' inductive biases (locality and translation equivariance), Dosovitskiy et al. [42] pioneered the adaptation of this paradigm to the visual domain with their vision transformer (ViT). Subsequently, Bandara and Patel [43] integrated a hierarchical Transformer encoder with an MLP decoder to capture multiscale context, while SwinSUNet [44] adopted a pure Siamese U-shaped Transformer architecture featuring a deep 32-block hierarchy. Nevertheless, pure Transformers face critical bottlenecks: they are data-hungry and computationally expensive, while their patch-based nature struggles to preserve fine-grained local details, often yielding imprecise boundaries compared to CNNs.

To combine the strengths of both frameworks, Chen et al. [45] pioneered a hybrid model, BIT, that utilizes a Transformer encoder to capture global context from CNN features via SelfAtt. Subsequently, hybrid architectures have increasingly emerged as the dominant approach in the field of CD [46], [47]. Yang et al. [46] introduced ConvFormer to enhance scene adaptability via parallel convolution and SelfAtt, whereas GCFormer [48] leverages a Multireceptive-Field Convolutional Attention mechanism to extract global context across varying scales [49], [50]. While achieving promising results, these hybrid architectures often suffer from inflated computational costs due to the stacking of complex modules. Furthermore, although some attempts have been made to bridge the semantic gap, they typically lack an efficient and lightweight fusion strategy. Crucially, most existing methods focus solely on enhancing feature extraction capacity, failing to establish a specialized framework to systematically decouple authentic changes from background noise. Therefore, a more efficient and systematic framework addressing the dual challenges of noise decoupling and the semantic gap is highly desired.

B. Spatial-Domain-Based Feature Interaction

A fundamental characteristic that distinguishes CD from other dense prediction tasks is the necessity of modeling interactions between bi-temporal features. The importance of feature interaction, well-documented across both homogeneous and heterogeneous data, is especially pertinent to CD due to the domain gap caused by variations in imaging conditions [51]. Numerous approaches have been proposed to address this challenge. Feng et al. [18] devised an intertemporal joint-attention block to manipulate the global feature distribution of each input, thereby stimulating information coupling between intralevel representations. Fang et al. [19] advocated for early feature interaction via a coattention mechanism, demonstrating the critical importance of interaction before fusion by treating features directly as attention maps. Zhao et al. [52] designed the FMCD model to enhance detection in complex scenes by modeling feature context through a multilevel interaction module, yielding highly representative features. Despite their promise, such methods are confined to the spatial domain where semantic content and stylistic variations remain intricately intertwined.

C. Frequency-Domain-Based Feature Interaction

Recently, feature interaction investigations have pivoted toward the frequency domain. While approaches leveraging global transforms like discrete cosine transform [23] and fast Fourier transform [24] effectively encode spectral context, they are inherently plagued by the Gibbs phenomenon during feature decoupling. This inevitably induces ringing artifacts that compromise spatial boundary fidelity [53]. To mitigate this, alternative approaches have explored wavelet-based architectures. FDINet [27] utilizes Frequency Decoupling Interaction specifically to discriminate between different change types (e.g., appearance versus disappearance) but relies on static interaction strategies that lack adaptability to complex scenes. Similarly, WS-Net++ [20] employs a wavelet Siamese structure to capture spatial and frequency differences in parallel. Nevertheless, it treats these two domains as isolated branches that are only merged via shallow concatenation at the end, limiting the ability of frequency cues to refine spatial representations. More critically, existing methods have insufficiently probed the feasibility of establishing a collaborative mechanism between frequency-domain and spatial-domain feature interactions to fully unlock the potential of their combined strengths.

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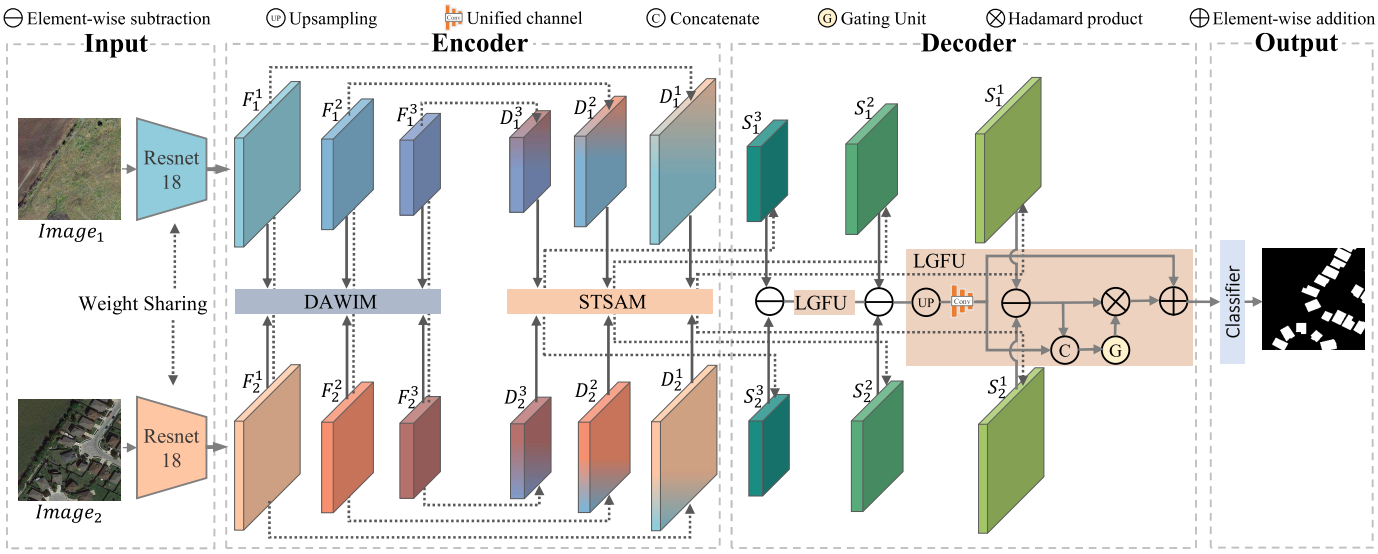


Fig. 2. Overall architecture of the proposed FSG-Net, where F_i^j, D_i^j, and S_i^j represent feature maps from the backbone, DAWIM, and STSAM, respectively. The subscript $i \in \{1, 2\}$ denotes the temporal phase, while the superscript $j \in \{1, 2, 3\}$ indicates the feature scale. Following the FCN paradigm, the encoding stage comprises feature extraction and enhancement via DAWIM and STSAM, while the decoding stage is responsible for spatial resolution reconstruction using LGFU.

To remedy these deficiencies, we propose a synergistic paradigm where frequency-domain interaction serves as a prefiltering stage to suppress pseudochanges, explicitly guiding the subsequent spatial interaction to enhance genuine changes.

III. Methodology

A. Method Overview

Algorithm 1 FSG-Net CD Process

Input: A pair of registered images $\{I_1, I_2\}$
Output: A predicted change map P
// Step 1: Extract multiscale backbone features
$\{F_1^j\}_{j=1}^3 \leftarrow \text{Backbone}(I_1)$
$\{F_2^j\}_{j=1}^3 \leftarrow \text{Backbone}(I_2)$
// Step 2: Refine features with Frequency-Spatial Synergy
for $j = 1 \rightarrow 3$ **do**
  $\{D_1^j, D_2^j\} \leftarrow \text{DAWIM}(F_1^j, F_2^j)$ // Suppress pseudochanges in Frequency domain
  $\{S_1^j, S_2^j\} \leftarrow \text{STSAM}(D_1^j, D_2^j)$ // Enhance genuine changes in Spatial domain
end for
// Step 3: Harmonize difference features with LGFU
$\Delta S_{\text{fused}}^3 \leftarrow |S_1^3 - S_2^3|$ // Initialize fused feature
for $j = 3 \rightarrow 2$ **do**
  $\Delta S^{j-1} \leftarrow |S_1^{j-1} - S_2^{j-1}|$
  $\Delta S_{\text{fused}}^{j-1} \leftarrow \text{LGFU}(\Delta S_{\text{fused}}^{j}, \Delta S^{j-1})$
end for
// Step 4: Generate the final change map
$P \leftarrow \text{PredictionHead}(\Delta S_{\text{fused}}^1)$

The overall architecture of FSG-Net, illustrated in Fig. 2, is designed as a principled, cascaded pipeline that systematically mitigates background noise and semantic gaps. Initially, the DAWIM suppresses pseudochanges by leveraging wavelet-based decoupling and adaptive filtering mechanisms in the frequency domain. Complementing this, the STSAM enhances the saliency of authentic change regions via global–local attention in the spatial domain. Finally, the LGFU exploits high-level semantics to selectively gate and fuse shallow details, ensuring precise boundary delineation. The pseudocode of FSG-Net is in Algorithm 1.

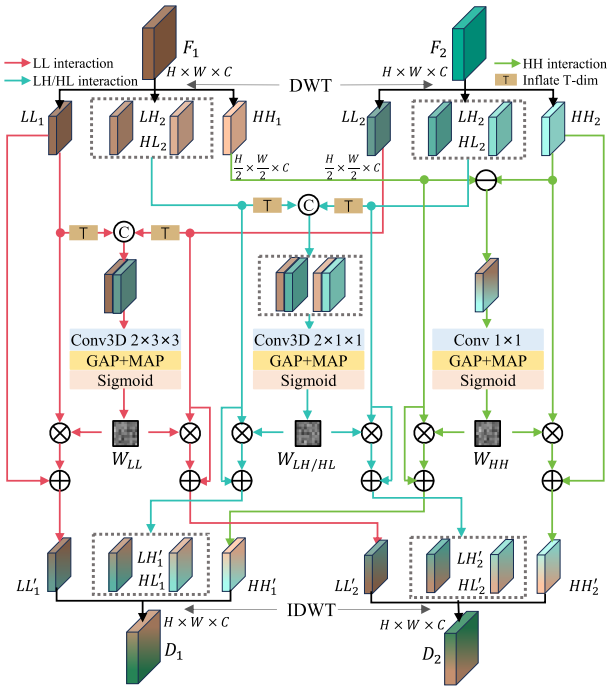


Fig. 3. Operational workflow of the DAWIM.

B. Discrepancy-Aware Wavelet Interaction Module

While the principle of frequency-domain interaction is promising, the key challenge lies in how to process the different frequency components to simultaneously enhance change signals while suppressing noise. The Wavelet Trans-

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form offers advantages such as time-frequency localization and multifrequency subband analysis, making it particularly well-suited for CD tasks involving scale-sensitive targets [20]. Accordingly, the DAWIM employs distinct processing strategies tailored to the specific characteristics of each frequency component. The operational workflow of this module is illustrated in Fig. 3.

Specifically, a 2-D Haar discrete wavelet transform (DWT) is first applied to decompose the bi-temporal spatial feature maps, denoted as F_i $(i \in \{1, 2\})$, into four frequency-domain subcomponents, denoted as $\{LL_i, LH_i, HL_i, HH_i\}$. This standard decomposition can be represented as

$$\{LL_i, LH_i, HL_i, HH_i\} = \mathrm{DWT}(F_i) \tag{1}$$

where DWT utilizes separable Haar wavelet filters. Subsequently, each frequency component undergoes an appropriate processing strategy based on its characteristics to maximize the benefits of frequency feature interaction.

The low-frequency components LL_i primarily represent large-scale spatial patterns and gradual radiometric variations, such as land cover distribution and global illumination trends [54]. Meanwhile, they exhibit relatively stable changes in the time domain, primarily reflecting gradual background variations [21]. To capture these slow temporal dynamics effectively, a 3-D convolution is applied to extract and then enhance the temporal features embedded within these frequency bands. Specifically, a temporal dimension is first inflated to $LL_i \in \mathbb{R}^{C\times H\times W}$, forming $\hat{LL}_i \in \mathbb{R}^{C\times 1\times H\times W}$. Then, $\hat{LL}_1$ and $\hat{LL}_2$ are concatenated along the temporal dimension to form $LL \in \mathbb{R}^{C\times 2\times H\times W}$ and a 3-D convolution is applied to enhance the temporal relationship modeling between the bi-temporal features. The kernel size is set to $2 \times 3 \times 3$, to perform feature fusion along the temporal dimension, while capturing spatial context from a 3×3 neighborhood. Finally, a reshaping operation is performed on the features $\hat{LL} \in \mathbb{R}^{C\times 1\times H\times W}$ obtained from the 3-D convolution to embed the temporal dynamics into the channels, resulting in low-frequency interaction features $\widetilde{LL} \in \mathbb{R}^{(C\times 1)\times H\times W}$. The entire operation can be formulated as

$$\widetilde{LL} = \mathrm{Reshape}\left(\mathrm{Conv3D}_{2\times 3\times 3}\left(\mathrm{Concat}\left[\hat{LL}_1, \hat{LL}_2\right]\right)\right) \tag{2}$$

where Concat represents concatenation along the temporal dimension.

Compared to the LL_i, the mid-frequency components (LH_i/HL_i) encapsulate more localized temporal variations, particularly abrupt structural changes in the horizontal and vertical directions [55]. We, therefore, adopt an interaction method similar to that used for components LL_i, using 3-D convolution to form $\widetilde{LH}$ and $\widetilde{HL}$, but set the kernel size to $2 \times 1 \times 1$ to fuse features along the temporal dimension without compromising the spatial structure. Taking $\widetilde{LH}$ as an example

$$\widetilde{LH} = \mathrm{Reshape}\left(\mathrm{Conv3D}_{2\times 1\times 1}\left(\mathrm{Concat}\left[\hat{LH}_1, \hat{LH}_2\right]\right)\right). \tag{3}$$

The high-frequency component HH_i, typically associated with edges, contours, and fine-grained structural details, is processed using element-wise subtraction to obtain the difference feature [20], formulated as follows:

$$\widetilde{HH} = \delta\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 1}\left(|HH_1 - HH_2|\right)\right)\right). \tag{4}$$

After obtaining these frequency interaction features, we introduce an adaptive weighting mechanism based on modified squeeze-and-excitation networks (SENet) [56]. This approach dynamically enhances the saliency of change regions while mitigating the impact of irrelevant background noise. Taking $\widetilde{LL}$ as an example, both max pooling (MAP) and global average pooling (GAP) are applied to the $\widetilde{LL}$ simultaneously—the former highlights the most significant feature responses, while the latter ensures that the model does not focus solely on extreme variations but instead learns more robust features. Then, a global feature vector, obtained by concatenating the max-pooled and average-pooled features, is used to compute the channel-adaptive weight $W_{\widetilde{LL}}$. Finally, the computed weight is applied to the original LL_i component via the Hadamard product, followed by a residual connection. This operation aims to leverage change-aware features, derived from bi-temporal interaction, to guide and refine the original, mono-temporal representations. The formulation is described as follows:

$$W_{\widetilde{LL}} = \sigma\left(W_2\delta\left(W_1\left[\mathrm{GAP}\left(\widetilde{LL}\right), \mathrm{MAP}\left(\widetilde{LL}\right)\right]\right)\right) \tag{5}$$

$$LL_i' = W_{\widetilde{LL}} \odot LL_i + LL_i. \tag{6}$$

Here, σ is the Sigmoid activation function, δ is the ReLU activation unit, and W_1, W_2 are the fully connected layer weights, where W_1 reduces dimensionality and W_2 restores it. The symbol $\odot$ denotes the Hadamard product.

Following the interaction between different frequency components, the feature change maps are reconstructed to the spatial domain through the application of the inverse discrete wavelet transform (IDWT). It can be formulated as

$$D_i = \mathrm{IDWT}\left(LL_i', LH_i', HL_i', HH_i'\right). \tag{7}$$

C. Synergistic Temporal–Spatial Attention Module

Following the frequency-domain feature interaction, noise interference induced by acquisition conditions is effectively attenuated. Our subsequent objective is to leverage spatial feature interaction to establish an information conduction pathway between the bi-temporal features, thereby mutually steering the allocation of attention to regions with genuine changes. To accomplish this, the STSAM is constructed by integrating a cross-attention (CrossAtt) augmented with temporal embeddings and a CoordAtt [17]. Compared to the SelfAtt, the CrossAtt equally processes the features of both temporal instances, avoiding bias toward a single image while simultaneously capturing temporal differences and global changes. Complementing this global temporal interaction, the CoordAtt is further employed to refine the local spatial structure, ensuring that fine-grained details, such as changes in building boundaries, are given adequate attention.

As shown in Fig. 4, $D_i \in \mathbb{R}^{C\times H\times W}$ are fed into a parallel processing stage consisting of two branches: the CrossAtt (branch 1) and the CoordAtt (branch 2). In branch 1, learning

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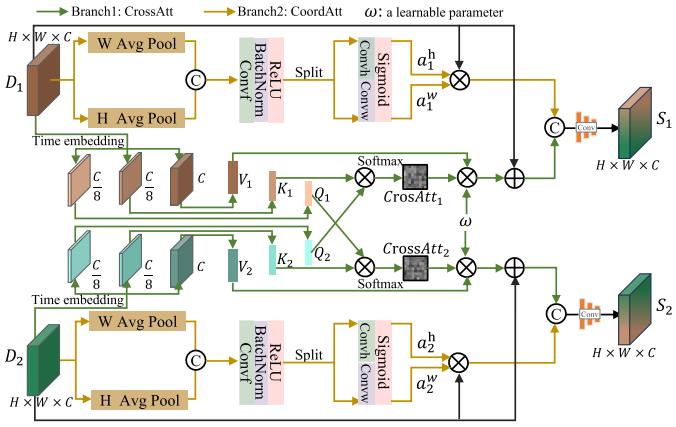


Fig. 4. Internal mechanism of the STSAM.

temporal embeddings (inspired by position embeddings in Transformer) [41], denoted as $T_i \in \mathbb{R}^{C\times 1\times 1}$, are first introduced and then added to D_i through Python's broadcasting mechanism, aiming to amplify the temporal discrepancies between inputs. Next, the enhanced features are projected into query (Q_i), key (K_i), and value (V_i) spaces using shared 1×1 convolutions. To maintain computational efficiency and in line with common practice, the channel dimensions of Q and K are compressed to 1/8 of the original, while the dimension of V is preserved.

Subsequently, the Q from the counterpart is used to perform similarity computation and weighting operations within its own $K - V$ space, ultimately yielding the final CrossAtt output. This output is then scaled by a learnable parameter ω and added back to the original feature D_i via a residual connection. The process is described by the following formulas:

$$Q_i, K_i, V_i = \text{Linear}\left(\text{Conv}_{1\times 1}\left(D_i + T_i\right)\right) \tag{8}$$

$$CrossAtt_i = \text{Softmax}\left(Q_j \cdot K_i^T\right) \cdot V_i \tag{9}$$

$$D_i^{\text{branch1}} = \omega \cdot CrossAtt_i + D_i \tag{10}$$

where $i, j \in \{1, 2\}$ are temporal indices with $i \neq j$, and ω is a learnable scaling parameter initialized to 0 for training stability.

In branch 2, adaptive average pooling is first applied to $D_i \in \mathbb{R}^{C\times H\times W}$ along the height and width directions, resulting in two directional feature map tensors: $x_i^h \in \mathbb{R}^{C\times H\times 1}$ and $x_i^w \in \mathbb{R}^{C\times 1\times W}$ (after dimension transposition). Then, x_i^h and x_i^w are concatenated along the spatial dimension and processed through a 1×1 convolution to fuse the pooled information from both directions. Next, the fused features are split, convolved, and passed through a sigmoid activation to generate the CoordAtt coefficients $a_i^h \in \mathbb{R}^{C\times H\times 1}$ and $a_i^w \in \mathbb{R}^{C\times 1\times W}$ for the height and width directions, respectively. Finally, the input feature D_i is reweighted through a Hadamard product with the computed attention coefficients a_i^h and a_i^w to perform the weighting operation

$$x_i^h = \text{AvgPool}_w\left(D_i\right), \quad x_i^w = \text{AvgPool}_h\left(D_i\right) \tag{11}$$

$$f_i = \delta\left(\text{BN}\left(\text{Conv}_f\left(\text{Concat}\left[x_i^h, \left(x_i^w\right)^T\right]\right)\right)\right) \tag{12}$$

$$D_i^{\text{branch2}} = D_i \odot \sigma\left(\text{Conv}_h\left(f_i^h\right)\right) \odot \sigma\left(\text{Conv}_w\left(f_i^w\right)\right) \tag{13}$$

where AvgPool_w and AvgPool_h are 1-D average pooling operations. Concat denotes concatenation along the spatial dimension, Conv_f, Conv_h, and Conv_w are separate 1×1 convolutions, δ is ReLU, and σ is Sigmoid. The fused tensors f_i are divided into f_i^h and f_i^w to generate directional attention maps.

The resultant feature maps from both pathways are then integrated through channel-wise concatenation and a subsequent convolutional fusion, yielding the fused output representation

$$S_i = \text{Conv}_{1\times 1}\left(\text{Concat}\left[D_i^{\text{branch1}}, D_i^{\text{branch2}}\right]\right). \tag{14}$$

Subsequently, the multiscale difference features ΔS^j are obtained by performing element-wise subtraction on the S_1^j and S_2^j. The superscript $j \in \{1, 2, 3\}$ indicates the feature scale.

D. Lightweight Gated Fusion Unit

After the previous two steps, the subsequent challenge lies in effectively fusing them to generate a final change feature map that incorporates both deep semantic information and shallow spatial details. Deliberately bypassing the intricate designs in feature decoders, we introduce a LGFU that establishes cross-scale information "flow" to aggregate both global context and local details. The fundamental principle of the LGFU is to selectively fuse features in the decoder's upsampling path. Instead of naively merging high-level semantic features with all available low-level details—which are often contaminated by noise and irrelevant clutter—the LGFU introduces a learnable gating mechanism. This gate adaptively controls the proportion of shallow-feature information that flows through at a pixel-wise level, effectively filtering out noise while preserving the integrity of meaningful change boundaries.

Specifically, given the jth layer's deep difference feature $\Delta S^j \in \mathbb{R}^{C_j\times(H/2^j)\times(W/2^j)}$, it first undergoes $2\times$ upsampling and a 1×1 convolution to align with the adjacent difference feature, $\Delta S^{j-1} \in \mathbb{R}^{C_{j-1}\times(H/2^{j-1})\times(W/2^{j-1})}$, in terms of spatial resolution and channel dimension, respectively. Subsequently, the aligned deep features and the shallow features are concatenated along the channel dimension and then fed into the gating unit (GU). The GU consists of a 3×3 convolution, followed by Batch Normalization, a ReLU activation, a 1×1 convolution, and a terminal Sigmoid function, which outputs a single-channel gating map, $G_{map} \in \mathbb{R}^{1\times(H/2^{j-1})\times(W/2^{j-1})}$, with pixel values normalized to the range $[0, 1]$. The gating map G_{map} then modulates the shallow features ΔS^{j-1} pixel-wise by performing a Hadamard product, where the single-channel G_{map} is broadcast across all channels of ΔS^{j-1} via Python's broadcasting mechanism. This operation "gates" the model to learn at each spatial location how much fine-grained detail from the shallow layer should be passed through to refine the semantic predictions from the deep layer. Ultimately, the gated shallow features ΔS^{j-1} are incorporated into the aligned deep features ΔS_{up}^j using a residual-like addition. This can be expressed as

$$\Delta S_{\text{up}}^j = \text{Conv}_{1\times 1}\left(\text{Upsample}\left(\Delta S^j\right)\right) \tag{15}$$

$$G_{map} = \text{GU}\left(\text{Concat}\left[\Delta S_{\text{up}}^j, \Delta S^{j-1}\right]\right) \tag{16}$$

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$$\Delta S_{\text{fused}}^{j-1} = G_{map} \odot \Delta S^{j-1} + \Delta S_{\text{up}}^{j}. \tag{17}$$

This fusion process is iteratively applied in a bottom-up manner across the decoder, progressively integrating features from shallower layers until the final high-resolution change map is generated.

Finally, the entire network is optimized end-to-end using a composite loss function. To address the pervasive class imbalance in CD, we combine the Binary Cross-Entropy ($\mathcal{L}_{\text{BCE}}$) and Dice ($\mathcal{L}_{\text{Dice}}$) losses, leveraging the former's ability to enforce pixel-level accuracy and the latter's robustness to skewed distributions [20], [29]. The final objective function is a linear combination of the two

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}. \tag{18}$$

IV. Experiments and Analysis

A. Datasets

Comparative experiments were conducted on three publicly available datasets.

1) CDD [57] dataset, captured from Google Earth, contains 11 pairs of raw images with varied resolutions (0.03 to 1 m/pixel), comprising both seasonal and disaster-induced changes. This dataset is particularly challenging due to complex scenarios involving shadows, lighting variations, and cloud cover [58]. We used the official preprocessed version, which consists of 10 000, 3000, and 3000 image patches of size 256 × 256 for the training, validation, and test sets, respectively.
2) GZ-CD [59] dataset, sourced from Google Earth via BIGEMAP, comprises season-varying, high-resolution (0.55 m/pixel) image pairs of Guangzhou's suburbs. The original images were cropped into 256 × 256 nonoverlapping patches, resulting in 1073 clips (originally 1067 in the source paper) after filtering out those containing changed pixels. The final training, validation, and test sets were split with 751, 214, and 108 pairs, respectively.
3) LEVIR-CD [60] dataset, a large-scale benchmark for building CD, consists of 637 image pairs (0.5 m/pixel) sized at 1024 × 1024. These images document diverse building changes, such as the emergence of villas and warehouses, over a 5 to 14-year span in multiple cities across Texas, USA. Following the official protocol, the images are partitioned into 7120, 1024, and 2048 patches of 256 × 256 for training, validation, and testing, respectively.

B. Experimental Setup

1) Implementation Details: All experiments were conducted in PyTorch on a single NVIDIA RTX 4090 GPU. We optimized the network using the AdamW optimizer with a weight decay of 0.01. The initial learning rate was set to 1e-3 for the randomly initialized decoder and head, and 1e-4 for the pretrained backbone, both decayed to 1e-6 over 100 epochs via a cosine annealing schedule. A batch size of 32 was used for all datasets.

2) Evaluation Metrics: Quantitative evaluation is conducted using five standard metrics: Precision (Pre.), Recall (Rec.), $F1$-score ($F1$), Intersection over Union (IoU), and overall accuracy (OA)

$$\text{Pre.} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{Rec.} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$F1 = \frac{2 \times \text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{21}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{22}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{23}$$

where TP, FP, TN, and FN represent the counts of true positives, false positives, true negatives, and false negatives. We prioritize the $F1$ and IoU for providing a more balanced evaluation.

C. Comparisons With SOTAs

To validate the effectiveness and efficiency of FSG-Net, widely adopted SOTA methods used as benchmarks for bi-temporal CD are implemented and compared, including the following.

1) SNUNet [39] leverages a Siamese Nested U-Net architecture with channel attention and deep supervision to robustly extract multiscale change features.
2) BIT [45] pioneers a hybrid CNN-Transformer architecture, where a Transformer encoder leverages SelfAtt on CNN feature to capture long-range dependencies.
3) ChangeFormer [43] employs a Siamese hierarchical Transformer encoder and an MLP decoder to capture multiscale, long-range contextual details.
4) DMINet [18] constructs an intertemporal joint-attention module for early feature interaction, enabling guided, dual-branch difference acquisition.
5) AMTNet [10] utilizes an integrated CNN-Transformer for feature exchange-based domain alignment and attention-driven multiscale feature enhancement.
6) WS-Net++ [20] introduces a wavelet Siamese network that uses semi-supervised domain adaptation to mitigate appearance shifts and reduce pseudochanges.
7) ChangeMamba [61] adapts the Mamba architecture into a Siamese U-Net framework, enabling efficient global interaction for identifying semantic changes.
8) CDNeXt [47] proposes a Temporospatial Interactive Attention to explicitly model and correct for geometric perspective rotation and temporal style differences.
9) FTransDF-Net [26] features a dual architecture with a gated module for multiscale detail fusion and a frequency Transformer for long-range dependency capture.
10) ConvFormer [46] integrates parallel convolution, SelfAtt, and a novel Temporal Attention mechanism for difference-guided cross-temporal relationship modeling.

Above compared models were benchmarked using their publicly available implementations and default parameters.

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TABLE I

COMPARISON RESULTS ON THE THREE CD DATASETS. THE BEST SCORE IS HIGHLIGHTED IN BOLD. ALL RESULTS ARE DESCRIBED AS PERCENTAGES (%)

Methods	Ref.	CDD					GZ-CD					LEVIR-CD				
		Pre.	Rec.	F1	IOU	OA	Pre.	Rec.	F1	IOU	OA	Pre.	Rec.	F1	IOU	OA
SNUNet [39]	GRSL21	90.54	86.61	88.53	79.42	97.26	82.76	80.55	81.64	68.98	90.56	89.31	89.56	89.43	80.88	98.89
BIT [45]	TGRS21	90.67	86.44	88.50	79.37	97.26	82.90	80.68	81.77	69.16	90.63	89.24	89.37	89.30	80.67	98.88
ChangeFormer [43]	IGARSS22	91.54	89.31	90.41	82.50	97.68	84.59	84.28	84.43	73.06	91.90	91.97	89.03	90.48	82.62	99.02
DMINet [18]	TGRS23	92.45	90.72	91.58	84.47	97.96	89.04	86.52	87.76	78.19	93.71	92.27	88.94	90.57	82.77	99.03
AMTNet [10]	ISPRS23	92.32	92.91	92.61	86.24	98.19	88.51	85.27	86.86	76.77	93.28	91.34	89.76	90.54	82.72	99.02
WS-Net++ [20]	TGRS24	92.95	93.21	93.08	87.06	98.31	88.32	85.76	87.02	77.02	93.33	92.11	**90.05**	91.07	83.60	99.08
ChangeMamba [61]	TGRS24	92.84	93.41	93.12	87.13	98.31	89.32	87.17	88.23	78.94	93.94	91.76	89.54	90.64	82.88	99.03
CDNeXt [47]	JAG24	92.76	93.87	93.31	87.46	98.35	89.21	86.34	87.75	78.17	93.72	92.15	89.69	90.91	83.33	99.06
FTransDF-Net [26]	JAG25	93.10	94.03	93.56	87.90	98.42	89.13	87.45	88.28	79.02	93.95	92.19	89.73	90.94	83.39	99.07
ConvFormer [46]	TGRS25	**93.35**	93.19	93.28	87.41	98.36	89.11	87.58	88.33	79.11	93.97	92.14	89.69	90.90	83.32	99.06
FSG-Net	Ours	93.33	**95.01**	**94.16**	**88.96**	**98.56**	**90.85**	**88.20**	**89.51**	**81.01**	**94.61**	**92.53**	90.04	**91.27**	**83.94**	**99.10**

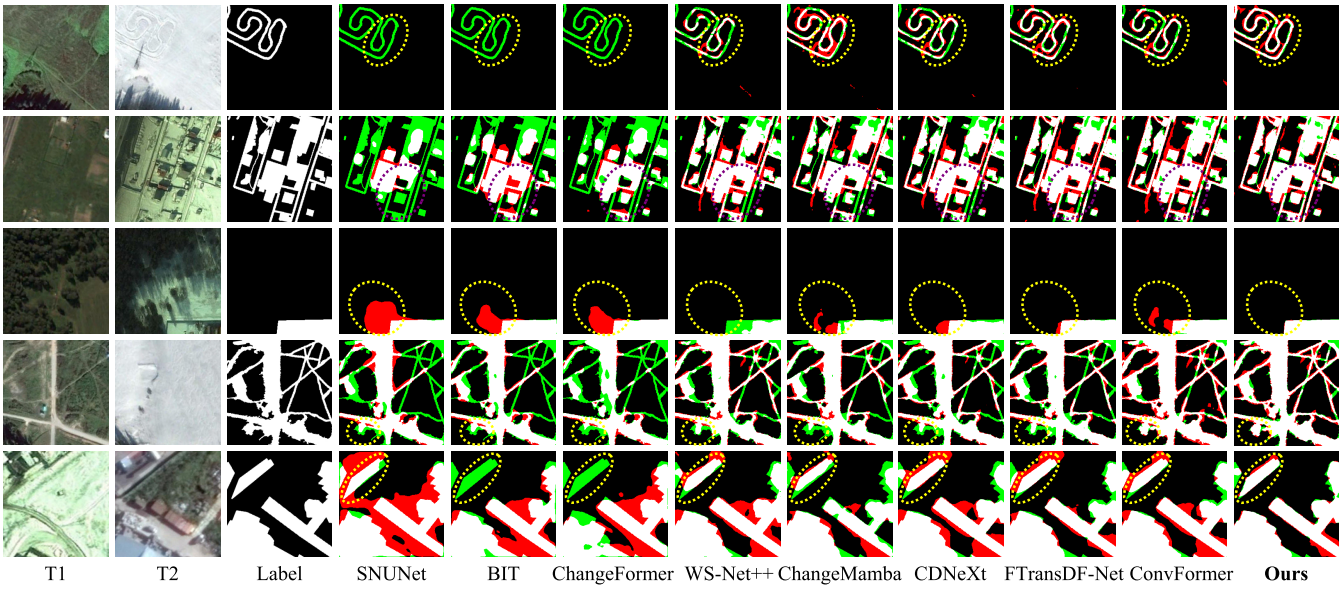


Fig. 5. Visualization of experimental results on the CDD dataset. White indicates true positives, black signifies true negatives, red marks false positives, and green highlights false negatives. Colored dashed boxes highlight the differences among the algorithms.

D. Experimental Results

1) Quantitative Evaluation: Table I summarizes the quantitative comparison on the CDD, GZ-CD, and LEVIR-CD datasets, where it is evident that the proposed FSG-Net attains superior results on nearly every metric.

Overall, FSG-Net achieves an $F1$ score of approximately 90% across all three datasets. A noticeable performance degradation for all methods is observed on the GZ-CD dataset. This is likely a consequence of two compounding factors: first, the data scarcity hinders the models from learning the complex change patterns robustly. Second, the dataset's high prevalence of seasonal discrepancies introduces significant ambiguity, further complicating the detection task with limited training data. Even under these demanding conditions, the proposed algorithm still exhibits strong performance, outperforming all comparison models with a Pre. of 90.85%, Rec. of 88.20%, $F1$ of 89.51%, IoU of 81.01%, and OA of 94.61%. This remarkable performance stems from the synergistic interplay between the DAWIM and STSAM. The former mitigates spurious variations by operating in the frequency domain, while the latter amplifies attention toward salient change regions. This collaborative mechanism results in a marked increase in the model's sensitivity to true changes. Likewise, the competitive performance of FSG-Net extends to the CDD and LEVIR-CD datasets, where it surpasses the recent ConvFormer by 1.55% and 0.62% in IoU, respectively.

2) Qualitative Evaluation: Figs. 5–7 present the predicted maps from a selection of representative methods across different datasets, with distinct colors indicating the detection results: TP (white), TN (black), FP (red), and FN (green). Colored dashed boxes highlight the differences among the algorithms.

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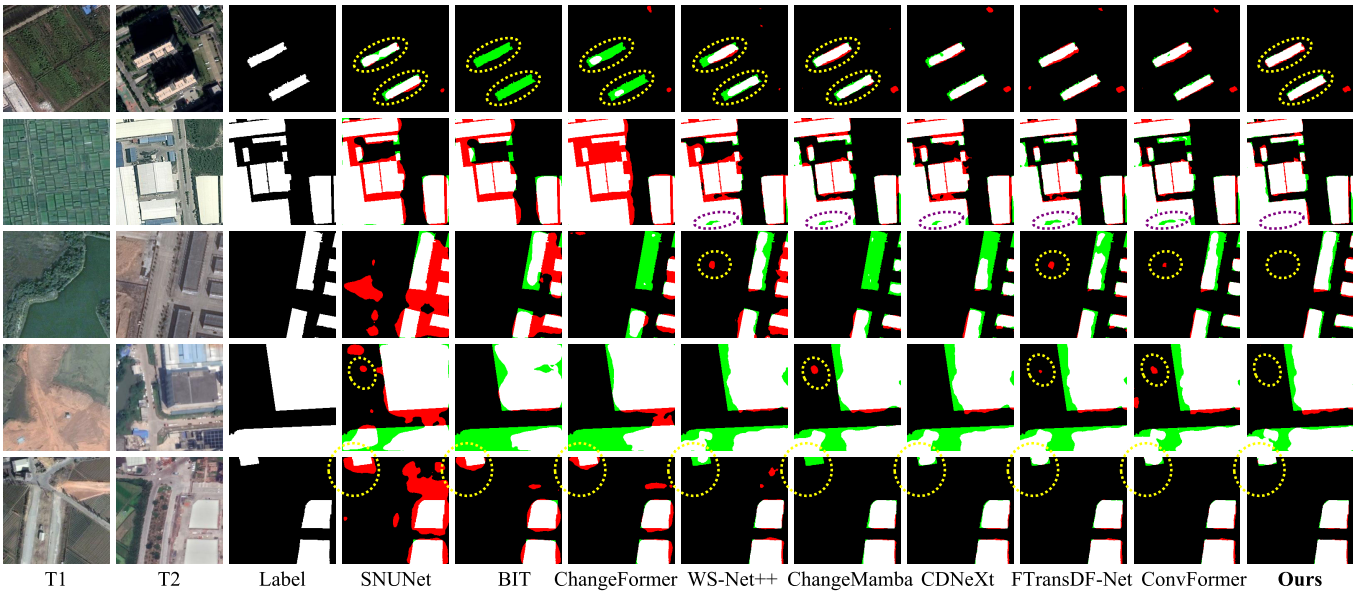


Fig. 6. Visualization of experimental results on the GZ-CD dataset. White indicates true positives, black signifies true negatives, red marks false positives, and green highlights false negatives. Colored dashed boxes highlight the differences among the algorithms.

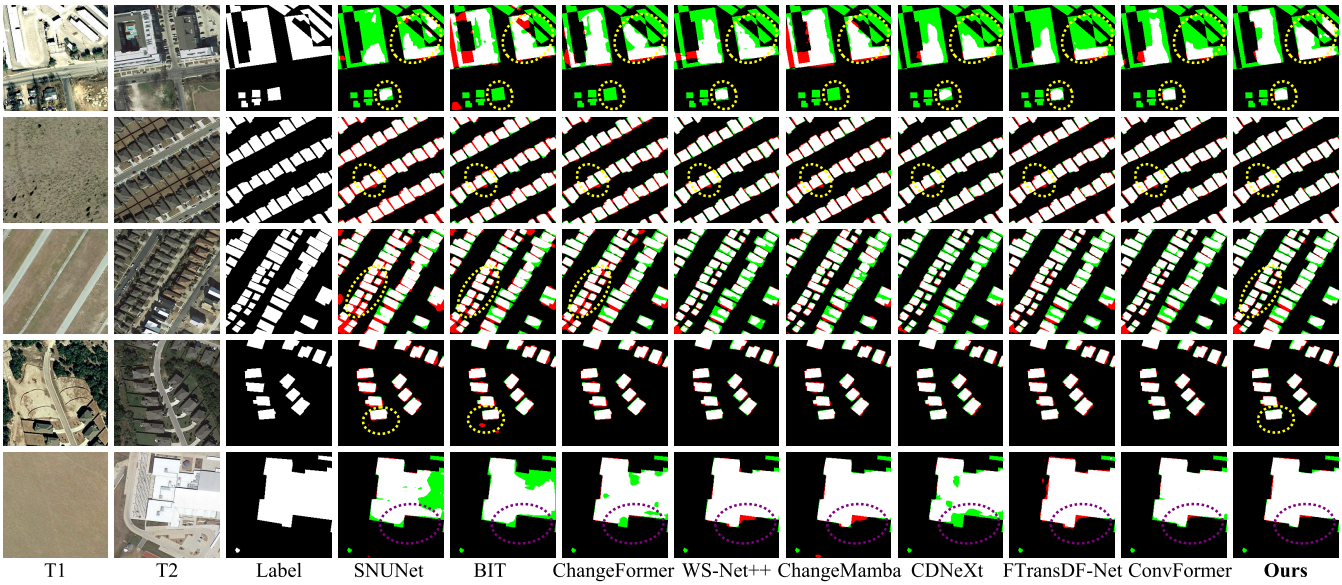


Fig. 7. Visualization of experimental results on the LEVIR-CD dataset. White indicates true positives, black signifies true negatives, red marks false positives, and green highlights false negatives. Colored dashed boxes highlight the differences among the algorithms.

Visualization on CDD (Fig. 5): As shown in the first and fourth rows, most algorithms struggle with the complex, crisscrossing network of rural paths, leading to either missed detections (e.g., BIT) or false alarms (e.g., ChangeMamba). Impressively, FSG-Net demonstrates exceptional resilience in this complex scene. It almost flawlessly traces the full extent of the path's change with high fidelity, notwithstanding a few negligible and inevitable misclassifications. In a particularly challenging scenario where illumination artifacts and seasonal variations co-exist (third row), severe shadows from arbor trees induce most algorithms to generate large, contiguous regions of pseudo-changes. This vulnerability is precisely what FSG-Net addresses through its differential frequency-domain interaction, which is designed to disentangle semantic changes from such nonsemantic, illumination-induced noise. Collectively, the aforementioned examples provide compelling evidence of FSG-Net proficiency in both suppressing pseudochanges and capturing genuine change targets.

Visualization on GZ-CD (Fig. 6): A closer look at the 1st row reveals that in the scenario where two new buildings are constructed on farmland, FSG-Net not only segments the entire building framework completely but also preserves the boundary integrity with remarkable clarity at the transition between the foreground (changed) and background (unchanged) regions. Such superior performance is intrinsically linked to the contribution of the STSAM. On one hand, the CrossAtt, augmented with temporal embeddings, models long-range contextual relationships across time between the

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TABLE II
ABLATION STUDY OF DIFFERENT MODULES. ALL RESULTS ARE DESCRIBED AS PERCENTAGES (%)

DAWIM	STSAM	LGFU	CDD		GZ-CD		LEVIR-CD	
			F1	IOU	F1	IOU	F1	IOU
Baseline			90.33	82.37	84.97	73.87	90.29	82.30
✓	×	×	91.97	85.13	86.80	76.68	90.45	82.57
×	✓	×	91.60	84.50	86.55	76.29	90.71	83.00
×	×	✓	91.15	83.74	85.80	75.13	90.62	82.85
×	✓	✓	92.50	86.05	87.73	78.14	91.05	83.57
✓	×	✓	92.88	86.71	87.95	78.49	90.73	83.03
✓	✓	×	93.35	87.53	88.57	79.48	91.01	83.50
Ours			**94.16**	**88.96**	**89.51**	**81.01**	**91.27**	**83.94**

bi-temporal features. On the other hand, the CoordAtt further enhances the focus on the fine-grained details of target edges. In this manner, changes, irrespective of their scale (global or local), are given due consideration and are not overlooked.

Visualization on LEVIR-CD (Fig. 7): The dataset, specifically labeled for building CD, is characterized by a wide spectrum of change scenes. It features changes ranging from the emergence of large-scale industrial structures, such as factories and warehouses, to the construction of dense residential clusters. The morphological heterogeneity of these structures presents a substantial hurdle for accurate boundary delineation. Owing to the LGFU, which narrows the semantic gap between high-level and low-level features, FSG-Net maintains the integrity of building structures and the sharpness of their edges more effectively than all competing algorithms. Consistent with the quantitative metrics in Table I, the qualitative analysis further underscores the SOTA capabilities of FSG-Net on the LEVIR-CD.

E. Ablation Experiment and Analysis

To provide a structured and meaningful evaluation, we benchmark against a competitive baseline, incorporating proven technologies like basic frequency differencing and Self-Att (replacing DAWIM and STSAM, respectively) [15], [27], [45]. The decoder fuses cross-scale difference features using a U-Net mechanism: upsampling followed by concatenation and a 1×1 convolution. This strong baseline serves to demonstrate that the performance gains of FSG-Net are not merely due to the inclusion of generic enhancements, but stem from specific, refined designs. Table II presents the ablation results for each component, focusing on the $F1$ and IoU.

1) Effectiveness of DAWIM: The primary role of the DAWIM is to suppress pseudo-changes by discerningly processing features in the frequency domain. To validate this, we isolated its effect by adding it to the baseline. As shown in Table II, this integration yields a consistent performance boost across all datasets. Most notably, it achieves $F1$ improvements of 1.64% on CDD and 1.83% on GZ-CD. The fact that these substantial gains occur on datasets characterized by significant seasonal variations strongly corroborates the exceptional capability of the DAWIM in mitigating task-irrelevant interference. This confirms that the strategy of applying

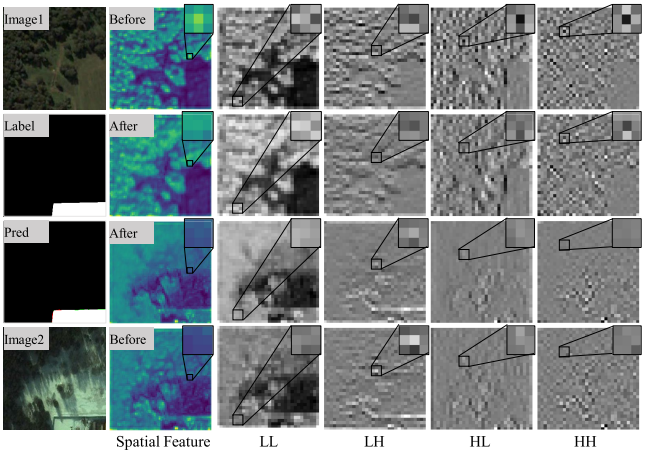


Fig. 8. Visualization of the DAWIM's feature modulation process. **Rows 1 & 4** show the features before interaction for Image1 and Image2. **Rows 2 & 3** show the features after interaction. The ground truth change mask (Label) and the model's final prediction (Pred) are also provided for reference. The columns display the input image, the spatial feature map, and its wavelet components.

distinct, tailored interactions to different frequency components is more effective than a simple, uniform differencing approach. Comparing the individual deployment of DAWIM and STSAM with their joint integration, it is evident that this combination yields a significant performance gain over their individual contributions. This reciprocal effect can be attributed to a sequential enhancement mechanism: DAWIM's initial filtering of background noise allows the STSAM to subsequently focus its temporal–spatial attention mechanism on authentic change targets with greater precision, thereby achieving a cumulative benefit greater than the sum of its parts.

To provide a more intuitive understanding of the DAWIM, as shown in Fig. 8, we visualize the normalized internal feature representations before and after the frequency-domain interaction. The figure is structured in a "sandwich" layout, where the features maps before interaction (Rows 1 and 4 for T1 and T2, respectively) flank after interaction (Rows 2 and 3). From the magnified insets highlighting feature differences before and after interaction, we can draw two key observations. First, the module effectively preserves crucial spatial structures, as evidenced by the overall consistency in feature patterns before and after the interaction. Second, it successfully introduces temporal dynamics, indicated by the subtle yet distinct changes in feature intensity and texture postinteraction, an effect that is quantitatively supported by the results in Table II.

2) Effectiveness of STSAM: The STSAM is designed to refine the representation of genuine changes by creating an information flow pathway between bi-temporal features to mutually steer the attention distribution effectively. As detailed in Table II, ablating the STSAM module leads to a drop in $F1$ of 1.28%, 1.56%, and 0.54% on the CDD, GZ-CD, and LEVIR-CD datasets, respectively, compared to the full model. By capturing temporal dependencies globally via CrossAtt augmented with temporal embedding and encoding precise spatial locations locally through CoordAtt, the STSAM

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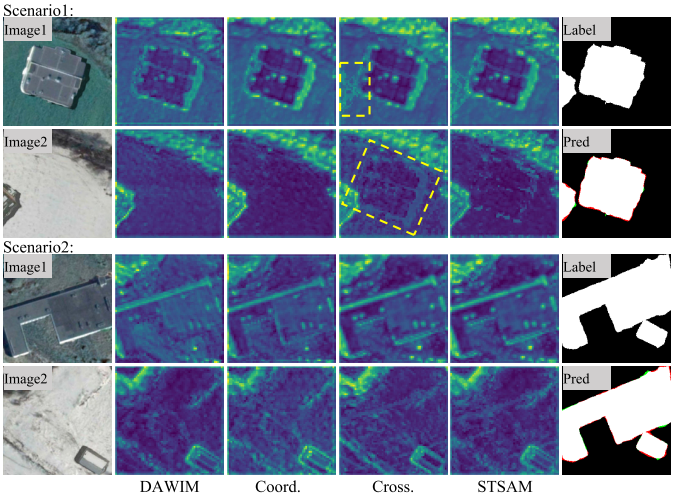


Fig. 9. Visualization of the progressive feature refinement process within the STSAM across two challenging scenarios. **Columns** from left to right show: the input images (Image1, Image2), and the feature heatmaps after the DAWIM, Coordinate Attention (Coord.), Cross-Attention (Cross.), and the final STSAM stage. For each scenario, the ground truth (Label) and the final prediction (Pred) are provided for comparison. The yellow boxes highlight the reciprocal attention focusing mechanism of the CrossAtt.

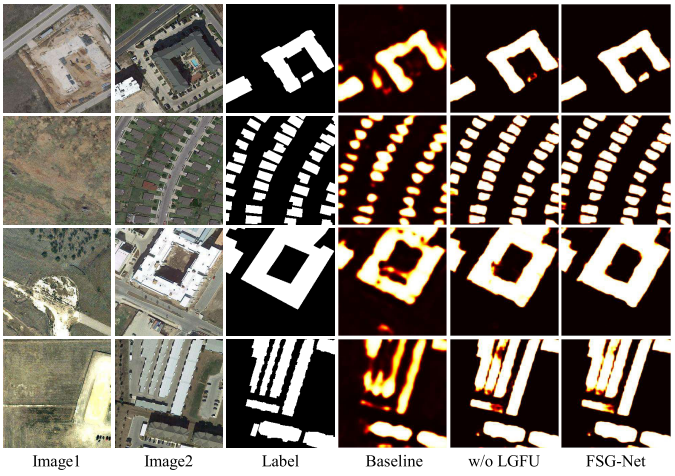


Fig. 10. Visualization of the LGFU's contribution to refining prediction confidence heatmaps. **Columns** from left to right show: the bi-temporal images (Image1, Image2), the ground truth (Label), and the predicted heatmaps from Baseline, the model without LGFU (w/o LGFU), and the complete FSG-Net. The color intensity corresponds to the predicted change probability, ranging from black (low) to white (high).

ensures that genuinely changed regions, irrespective of their scale or complexity, are saliently represented, thereby achieving complementary temporal–spatial focus on change targets.

Fig. 9 illustrates the progressive feature refinement by the STSAM in two representative scenarios featuring building changes amidst seasonal shifts. Evidently, the CoordAtt branch sharpens the feature representation, accentuating fine-grained spatial details and preserving the integrity of building boundaries. The yellow-boxed regions in the fourth column warrant special attention as they highlight a key observation: through the temporally-enhanced CrossAtt, the T1 features perceive the new building construction in the T2 image, prompting a heightened focus on the counterpart location within itself. Conversely, the T2 features identify the demolition in T1 and similarly amplify attention on their own counterpart region. The final fused feature map showcases the synergy of these two streams: it inherits the high-level semantic focus from the temporal branch while being refined with the high-fidelity structural information from the spatial branch. This fusion enables the model to robustly identify multiscale changes while preserving sharp boundary integrity, as evidenced in the final segmentation results. The bottom two rows present a similar yet more challenging case where a new building is camouflaged by snow cover, making it visually blend with the environment. Remarkably, due to the robust frequency-spatial interplay, FSG-Net precisely delineates the changed area despite these challenging camouflage-like artifacts, a result clearly visible in the final prediction.

3) Effectiveness of LGFU: Bridging the semantic gap between deep abstract features and shallow detail-rich features is another critical challenge for accurate CD. The LGFU achieves this by leveraging high-level semantic information to guide and enrich shallow features. A clear observation from Table II reveals the significant contribution of the LGFU. Integrating the LGFU individually into the baseline boosts the IoU by 1.37%, 1.26%, and 0.55% on the respective datasets. More revealingly, when LGFU is the sole module ablated from the full model, the performance degradation is even more pronounced, with IoU dropping by 1.43% and 1.53% on the first two datasets. This asymmetry is a crucial finding: It indicates that the LGFU's contribution is amplified when it operates on the highly refined features generated by DAWIM and STSAM. This validates a powerful synergistic effect within our architecture, where the modules not only contribute individually but also enhance each other's efficacy, culminating in a final model that is greater than the sum of its parts.

Fig. 10 visualizes the LGFU's contribution using a predicted probability heatmap, where the color of each pixel maps to the model's confidence score for the change class. The color intensity signifies the probability of change, ascending from black (lowest) through red and yellow to white (highest). It is easily observed that the baseline captures the general change regions, yet it struggles with blurry boundaries and noisy interferences. The predominantly yellow prediction map further suggests a lack of absolute certainty in its identification of changed areas. Only upon the final integration of the LGFU does the model produce exceptionally sharp and well-defined change boundaries that closely adhere to the ground truth. This provides compelling visual evidence that the LGFU is instrumental not just in bridging the semantic gap, but ultimately in translating high-level conceptual understanding into pixel-perfect boundary delineations.

4) Analysis on DAWIM's Interaction Strategies: To validate the internal design of DAWIM, we compared different frequency-domain interaction strategies, including: unified approaches (e.g., Difference, Conv2x1x1, Conv2x3x3) for all components, and the removal of the SE weighting mechanism and the residual connection, as detailed in Table III. Taking the CDD as an example, the DAWIM surpasses the other three unified strategies in $F1$ by 1.33%, 0.69%, and 0.51%, respectively, indicating that a tailored interaction strategy could maximize the benefits of frequency-domain feature interaction. Another noteworthy observation is that removing the residual

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TABLE III
ANALYSIS ON DAWIM'S INTERACTION STRATEGIES. THE METRIC IS $F1$ SCORE (%)

Strategy	Weighted	Residual	CDD	GZ-CD	LEVIR-CD
Difference	✓	✓	92.83	88.03	91.09
$\text{Conv}_{2\times1\times1}$	✓	✓	93.47	88.42	91.12
$\text{Conv}_{2\times3\times3}$	✓	✓	93.65	88.92	91.17
w/o SE	×	✓	93.97	89.28	91.20
w/o Res	✓	×	92.57	87.81	90.94
DAWIM	✓	✓	**94.16**	**89.51**	**91.27**

TABLE IV
COMPARISON ON STSAM'S DIFFERENT COMPONENTS. THE METRIC IS $F1$ SCORE (%)

Component	CDD	GZ-CD	LEVIR-CD
Cross-Attention	93.69	88.93	91.02
Coord-Attention	92.36	87.41	90.31
Self-Attention	92.88	87.95	90.73
Self & Coord-Attention	93.47	88.74	90.91
w/o time embedding	94.01	89.34	91.18
STSAM	**94.16**	**89.51**	**91.27**

connection leads to a catastrophic collapse, with the $F1$ dropping by 1.59%, 1.70%, and 0.33% on the three datasets, respectively. It can be attributed to a well-known consensus in deep learning: removing residual connections hinders model convergence and impedes the smooth flow of gradients and information [41]. This suggests that a robust and easily optimizable architecture is a prerequisite for fully unleashing the potential of advanced frequency-domain interaction strategies.

5) Comparison on STSAM's Components: Table IV presents an in-depth ablation study, comparing different standard attention mechanisms and the effect of removing the temporal embedding from the complete STSAM.

Substituting SelfAtt with CoordAtt slightly decreases detection performance, due to the latter's inability to effectively capture long-range dependencies. When the two attention mechanisms are combined, the global modeling capabilities and direction-aware feature encoding yield a certain degree of performance improvement. Nevertheless, the inherent functional overlap in this combination keeps the $F1$ lower by 0.22%, 0.19%, and 0.11% on the three datasets, respectively, compared to using only CrossAtt. The last two rows reveal that explicitly amplifying temporal disparities between bi-temporal features via time embedding can inject new life into the model, particularly when it hits a performance ceiling, with the $F1$ increasing by 0.15%, 0.17%, and 0.09%, respectively.

F. Discussion

Synthesizing the quantitative results (Table I) and qualitative evidence (Figs. 5–7), it is evident that FSG-Net effectively suppresses pseudo-changes and bridges the semantic gap through the orchestrated interplay of its DAWIM, STSAM, and LGFU modules. Demonstrating its superiority across diverse architectural paradigms, FSG-Net surpasses not only classic CNN-based models but also cutting-edge frequency-aware and Mamba-based competitors, securing top $F1$-scores of 94.16%, 89.51%, and 91.27% on the CDD, GZ-CD, and LEVIR-CD datasets, respectively. Providing intuitive evidence for this approach, the visualization results in Figs. 8 and 9 illustrate that DAWIM's frequency-domain interaction suppresses pseudo-changes, while STSAM's spatial interaction enhances genuine changes. This bidirectional interplay mitigates visual feature interference arising from disparate imaging conditions and promotes information coupling within intralevel representations. Moreover, the evidence in Fig. 10 substantiates the role of the LGFU. It leverages high-level semantics to guide the selection of valuable details from shallow features, resulting in the effective filtering of noise and the generation of clean, well-defined boundaries. However, the current architecture's robustness in extreme scenarios could be further enhanced. A notable limitation, illustrated in the final row of Fig. 7, is the failure to detect minute, isolated building changes amidst vast homogeneous backgrounds.

TABLE V
COMPARISONS OF MODEL ARCHITECTURE AND EFFICIENCY. THE CALCULATION INCLUDES THE BACKBONE

Method	Backbone	Frequency Interaction	Params(M)	FLOPs(G)
SNUNet	UNet++	×	12.04	54.82
BIT	ResNet18	×	3.55	4.35
ChangeFormer	MiT-b1	×	13.94	52.84
DMINet	ResNet18	×	6.24	14.42
AMTNet	ResNet18	×	16.44	14.71
WS-Net++	ResNet18	✓	–	–
ChangeMamba	VMamba-Tiny	×	17.13	45.74
CDNeXt	CNX-t	×	39.42	15.76
FTransDF-Net	DFG-UNet	✓	17.91	9.37
ConvFormer	w/ ST & BT-Blocks	×	37.72	5.14
Ours	ResNet18	✓	13.76	6.21

Table V compares the architectural complexity and efficiency. The model parameters (Params) and floating-point operations (FLOPs) were based on a unified image input size of $256 \times 256 \times 3$.

Beyond incorporating an advanced frequency interaction mechanism—a feature absent in most SOTA methods—FSG-Net achieves a superior balance in model complexity. While recent architectures like CDNeXt are parameter-heavy (39.42M) and ChangeFormer is computationally expensive (52.84G), FSG-Net achieves an exceptional tradeoff with only 13.76M parameters and 6.21 GFLOPs. This high efficiency is a direct result of our synergistic design philosophy, rather than relying on the brute-force stacking of complex components.

V. CONCLUSION

In this article, we have proposed FSG-Net to address two critical concerns in CD tasks: the high false alarm rates caused by pseudochanges from varying imaging conditions, and the semantic gap between deep, abstract features and shallow, detail-rich features. For the former, FSG-Net

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introduces a synergistic frequency-spatial separation strategy. More specifically, it leverages the DAWIM to optimize feature representations by applying tailored processing to different frequency-domain components, thereby suppressing interference from task-irrelevant factors. Subsequently, the STSAM module establishes an information pathway between the bitemporal features, which in turn mutually guides their attention toward genuine change targets. This synergistic "push–pull" mechanism is designed to maximize the benefits of feature interaction. To tackle the second challenge, the LGFU leverages high-level semantics to guide and filter shallow-level details, ensuring that change targets are well-represented regardless of their scale. Extensive experiments on three public datasets demonstrate that FSG-Net delivers SOTA performance, exhibiting a superior tradeoff between accuracy and efficiency.

For future research, our efforts will be directed toward lightweight architectures and integrating multimodal data or constructing knowledge graphs for rule-based constraints to further enhance the model's capabilities in extreme and complex scenarios [62].

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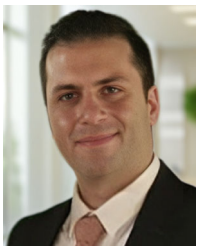
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